

WASP-43b

R-0361

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-43_260113 · created 2026-07-18T13:08:11+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2461053.9828 +/- 0.0021 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.197 +/- 0.019
Transit depth (Rp/Rs)^2	3.88 +/- 0.75 [%]
Orbital Inclination (inc)	80.01 +/- 1.41
Airmass coefficient 1 (a1)	1.048 +/- 0.015
Airmass coefficient 2 (a2)	-0.029 +/- 0.013
Scatter in the residuals of the lightcurve fit is	1.39 %
Best Comparison Star	#1 - [323, 154]
Optimal Aperture	3.06
Optimal Annulus	11.73
Transit Duration (day)	0.0451 +/- 0.0069

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.197 $\pm$ 0.019 vs 0.1594 (deviation 1.38 σ)
Duration fit vs published	0.0451 d vs 0.0512 d (deviation 0.62 σ)
Mid-time O-C	-2.5 min
Depth SNR	7.3
Residual scatter	1.39 %

Light curve

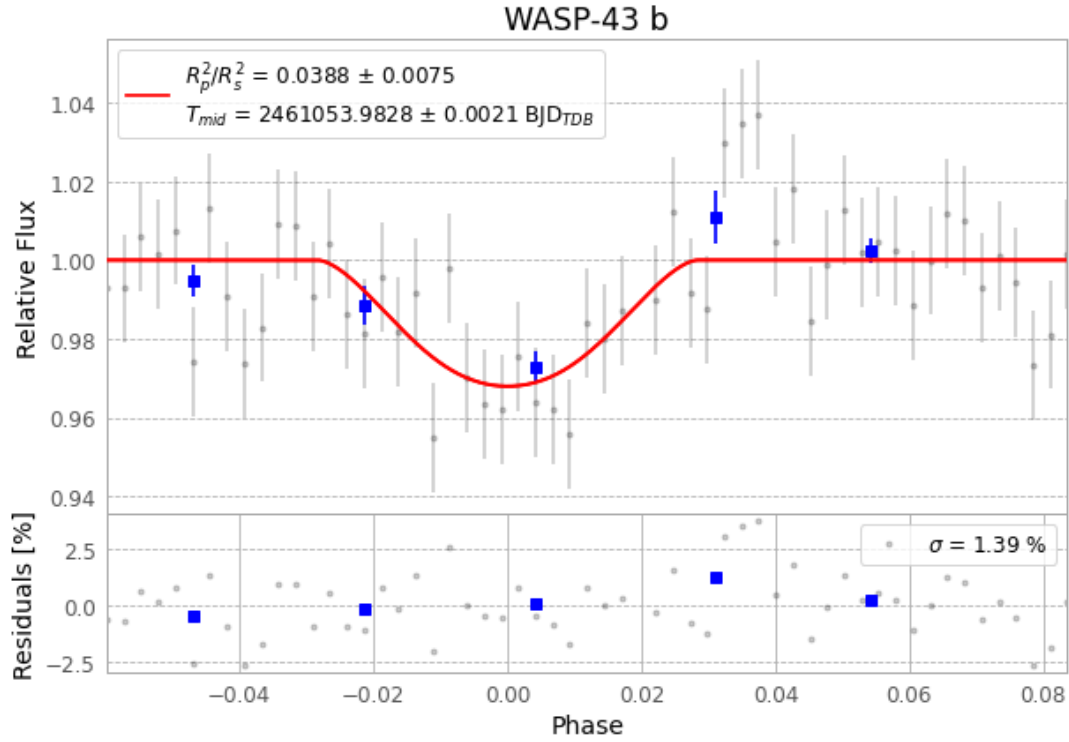


Figure 1 — FinalLightCurve_WASP-43 b_2026-01-13.png (SHA-256 1d46e0c3b8f7e944...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [443, 337], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 e73fcf404954f0c3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 9e597cf

Data provenance

57 FITS frames (37 MB), WASP-43260113101815.FITS → WASP-43260113130615.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-43-260113.log (d5e8e81540f2...), FinalLightCurve_WASP-43 b_2026-01-13.png (1d46e0c3b8f7...), FinalLightCurve_WASP-43 b_2026-01-13.csv (d636ec42cc3f...)

Compute resources

Wall time 100.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-18 13:08Z.